

The canonical four-corner mixed-norm sum has the Fatou property

Candidate full solution packet for arXiv:2006.01083

9 August 2026

Status: candidate full solution, likely valid pending expert review.

Abstract

For arbitrary sigma-finite measure spaces X and Y , consider

$$\mathcal{S} = L^1(X \times Y) + L^\infty(X \times Y) + L^{1,\infty}(X \times Y) + L^{\infty,1}(X \times Y)$$

with its canonical infimal-sum norm. Holighaus and Voigtlaender proved that this norm is equivalent to a Fatou norm, but left open whether the canonical norm itself has the Fatou property. We prove that it does. The proof also shows that the infimum defining the canonical norm is attained for every element of $\mathcal{S}$.

1 Source question

The source is N. Holighaus and F. Voigtlaender, *Schur-type Banach modules of integral kernels acting on mixed-norm Lebesgue spaces*, arXiv:2006.01083v2 (2020), later published in the *Journal of Functional Analysis* [1]. On page 30, in the unnumbered remark following Proposition A.7, the authors state that the canonical norm has the weak Fatou property, but that they “could not confirm whether it actually satisfies the Fatou property as well.”

The following proposition shows that the norm $F \mapsto \varrho_\otimes(|F|)$ is equivalent to the defining norm of the space $\mathbf{L}^1 + \mathbf{L}^\infty + \mathbf{L}^{1,\infty} + \mathbf{L}^{\infty,1}$.

Proposition A.7. *Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$ be σ -finite measure spaces. With $\varrho_\otimes$ as in Definition A.5 and $\|\bullet\|_{\mathbf{L}^1 + \mathbf{L}^\infty + \mathbf{L}^{1,\infty} + \mathbf{L}^{\infty,1}}$ as introduced in Definition 2.8, we then have*

$$\frac{1}{16} \cdot \|F\|_{\mathbf{L}^1 + \mathbf{L}^\infty + \mathbf{L}^{1,\infty} + \mathbf{L}^{\infty,1}} \leq \varrho_\otimes(|F|) \leq \|F\|_{\mathbf{L}^1 + \mathbf{L}^\infty + \mathbf{L}^{1,\infty} + \mathbf{L}^{\infty,1}}$$

for each measurable $F : X \times Y \rightarrow \mathbb{C}$.

Remark. In the terminology of solid function spaces, the proposition shows (in combination with Lemma A.6) that the space $\mathbf{L}^1 + \mathbf{L}^\infty + \mathbf{L}^{1,\infty} + \mathbf{L}^{\infty,1}$ with its canonical norm satisfies the *weak Fatou property*; see [33] § 65] for the definition. Here, we would like to remark that there is a characterization of (possibly infinite) families $(X_i)_{i \in I}$ of solid Banach spaces for which the sum $\sum_{i \in I} X_i$ with its natural norm satisfies the *Fatou property*; see [22]. This characterization, however, is quite technical, and we were unable to verify it in our setting. Thus, although we know that $\mathbf{L}^1 + \mathbf{L}^\infty + \mathbf{L}^{1,\infty} + \mathbf{L}^{\infty,1}$ with its natural norm satisfies the *weak Fatou property*, we could not confirm whether it actually satisfies the Fatou property as well.

Figure 1: The open statement on page 30 of arXiv:2006.01083v2.

Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$ be sigma-finite measure spaces. Following the source, for a measurable $F : X \times Y \rightarrow \mathbb{C}$ put

$$\|F\|_{p,q} = \left\| y \mapsto \|F(\cdot, y)\|_{L^p(\mu)} \right\|_{L^q(\nu)}, \quad p, q \in \{1, \infty\}.$$

Thus $L^{1,1} = L^1$ and $L^{\infty,\infty} = L^\infty$. The canonical norm in question is

$$\|F\|_{\mathcal{S}} = \inf_{F=F_{11}+F_{\infty\infty}+F_{1\infty}+F_{\infty 1}} (\|F_{11}\|_{1,1} + \|F_{\infty\infty}\|_{\infty,\infty} + \|F_{1\infty}\|_{1,\infty} + \|F_{\infty 1}\|_{\infty,1}). \quad (1)$$

2 Proof intuition

For $0 \leq F_n \uparrow F$, choose almost optimal four-way decompositions of each F_n . Solidity lets us make all four pieces positive, so every piece is bounded pointwise by F . After division by $1 + F$, the resulting vector of four pieces is bounded in a single Hilbert space L^2 built from an equivalent finite measure. Hilbert-space Cesaro compactness then produces almost everywhere limits for all four pieces simultaneously. The Fatou lemma in each of the four mixed-norm spaces preserves the four norm budgets, while the sums of the Cesaro means converge to F . Consequently the limiting pieces decompose F at cost no larger than $\lim \|F_n\|_{\mathcal{S}}$.

The only preliminary issue is that F might initially take the value infinity. Sigma-finiteness supplies bounded, strictly positive integrable functions on X and Y . Their tensor product belongs to all four associate spaces, so a duality estimate and monotone convergence force F to be finite almost everywhere.

3 The result

Theorem 1. *Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$ be sigma-finite measure spaces. The canonical norm (1) has the Fatou property. More precisely, if $0 \leq F_n \uparrow F$ almost everywhere, then*

$$\|F_n\|_{\mathcal{S}} \uparrow \|F\|_{\mathcal{S}},$$

with the convention that both sides may be infinite.

Moreover, for every $F \in \mathcal{S}$ there are

$$F_{11} \in L^1, \quad F_{\infty\infty} \in L^\infty, \quad F_{1\infty} \in L^{1,\infty}, \quad F_{\infty 1} \in L^{\infty,1}$$

whose sum is F and for which the infimum in (1) is attained.

We isolate two elementary compactness facts.

Lemma 2 (positive decompositions). *If $G \geq 0$ belongs to $\mathcal{S}$, then the infimum in (1) can be computed using only decompositions into four nonnegative functions. The sum norm is solid, and $\|F\|_{\mathcal{S}} = \| |F| \|_{\mathcal{S}}$.*

Proof. Suppose $G = \sum_{i=1}^4 U_i$ is any complex-valued decomposition and put $R = \sum_i |U_i|$. On $\{R > 0\}$ define

$$V_i = G \frac{|U_i|}{R},$$

and put $V_i = 0$ on $\{R = 0\}$. Since $G = |\sum_i U_i| \leq R$, we have $0 \leq V_i \leq |U_i|$ and $\sum_i V_i = G$. Solidity of each of the four component spaces shows that this replacement cannot increase any component norm. Taking infima proves the first assertion. Multiplying a positive decomposition of $|F|$ by the measurable phase of F proves the reverse comparison and hence $\|F\|_{\mathcal{S}} = \| |F| \|_{\mathcal{S}}$. The same argument with $0 \leq G \leq H$ proves solidity of the sum norm. $\square$

Lemma 3 (simultaneous Cesaro extraction). *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let (U_n) be bounded in $L^2(\mathbb{P}; \mathbb{R}^m)$, where $m < \infty$. There is a subsequence (U_{n_k}) such that the Cesaro means*

$$A_N = \frac{1}{N} \sum_{k=1}^N U_{n_k}$$

have a subsequence converging almost everywhere.

Proof. The closed linear span of (U_n) is a separable Hilbert space. A diagonal argument against an orthonormal basis gives a weakly convergent subsequence. After subtracting its weak limit, choose a further subsequence (W_k) with $|\langle W_j, W_k \rangle| \leq 2^{-k}$ for $j < k$. Expanding the square shows that the norms of $N^{-1} \sum_{k \leq N} W_k$ tend to zero. Thus the Cesaro means converge in L^2 . Choosing a subsequence for which the squared L^2 errors are summable and applying Chebyshev's inequality followed by Borel–Cantelli gives almost everywhere convergence. $\square$

Proof of Theorem 1. The assertion is immediate when $L := \sup_n \|F_n\|_{\mathcal{S}} = \infty$, so assume $L < \infty$. If $\mu \otimes \nu$ is the zero measure, there is nothing to prove, so assume that it is nonzero.

Step 1: a strictly positive common dual test. Sigma-finiteness gives bounded functions $a : X \rightarrow (0, \infty)$ and $b : Y \rightarrow (0, \infty)$ with $a \in L^1(\mu)$ and $b \in L^1(\nu)$. (For example, use disjoint finite-measure exhaustions and summable positive coefficients.) Set $H(x, y) = a(x)b(y)$. Then $H > 0$ almost everywhere, $H \in L^1$, and

$$\begin{aligned} \|H\|_{1,1} &= \|a\|_1 \|b\|_1, & \|H\|_{\infty,\infty} &= \|a\|_{\infty} \|b\|_{\infty}, \\ \|H\|_{1,\infty} &= \|a\|_1 \|b\|_{\infty}, & \|H\|_{\infty,1} &= \|a\|_{\infty} \|b\|_1. \end{aligned}$$

Mixed-norm Holder inequalities therefore give a constant $C_H < \infty$ such that

$$\int_{X \times Y} |G| H d(\mu \otimes \nu) \leq C_H \|G\|_{\mathcal{S}} \quad (G \in \mathcal{S}). \quad (2)$$

Indeed, pair the four summands respectively with $L^\infty, L^1, L^{\infty,1}, L^{1,\infty}$ and take the infimum over all decompositions. By monotone convergence,

$$\int F H = \lim_n \int F_n H \leq C_H L.$$

Since $H > 0$ almost everywhere, $F < \infty$ almost everywhere.

Step 2: almost optimal positive decompositions. For each n , Lemma 2 yields nonnegative functions $G_{i,n}$ in the four component spaces E_i , in the order

$$(E_1, E_2, E_3, E_4) = (L^1, L^\infty, L^{1,\infty}, L^{\infty,1}),$$

such that

$$F_n = \sum_{i=1}^4 G_{i,n}, \quad \|F_n\|_{\mathcal{S}} \leq \sum_{i=1}^4 \|G_{i,n}\|_{E_i} \leq \|F_n\|_{\mathcal{S}} + \frac{1}{n}. \quad (3)$$

In particular $0 \leq G_{i,n} \leq F_n \leq F$. Passing to a subsequence, we may assume simultaneously that

$$\|G_{i,n}\|_{E_i} \longrightarrow \alpha_i \quad (1 \leq i \leq 4).$$

Since $\|F_n\|_{\mathcal{S}} \uparrow L$, (3) gives

$$\sum_{i=1}^4 \alpha_i = L. \quad (4)$$

Normalize H so that $H d(\mu \otimes \nu)$ is a probability measure $\mathbb{P}$, and define

$$U_n = \frac{(G_{1,n}, G_{2,n}, G_{3,n}, G_{4,n})}{1 + F}.$$

Every coordinate of U_n lies between zero and one, so (U_n) is bounded in $L^2(\mathbb{P}; \mathbb{R}^4)$. Lemma 3, followed if necessary by a further subsequence, supplies integers $N_j \rightarrow \infty$ for which the Cesaro means

$$A_{i,N_j} := \frac{1}{N_j} \sum_{k=1}^{N_j} G_{i,n_k}$$

converge almost everywhere to nonnegative functions G_i .

Because $F_{n_k} \rightarrow F$ pointwise,

$$\sum_{i=1}^4 G_i = \lim_{j \rightarrow \infty} \frac{1}{N_j} \sum_{k=1}^{N_j} F_{n_k} = F \quad \text{almost everywhere.}$$

Each E_i has the Fatou property: this is the usual Fatou lemma, applied first in the inner variable and then in the outer variable for the two mixed spaces. Hence

$$\begin{aligned} \|G_i\|_{E_i} &\leq \liminf_{j \rightarrow \infty} \|A_{i,N_j}\|_{E_i} \\ &\leq \lim_{j \rightarrow \infty} \frac{1}{N_j} \sum_{k=1}^{N_j} \|G_{i,n_k}\|_{E_i} = \alpha_i. \end{aligned}$$

Together with (4), this yields

$$\|F\|_{\mathcal{S}} \leq \sum_{i=1}^4 \|G_i\|_{E_i} \leq \sum_{i=1}^4 \alpha_i = L.$$

Solidity gives $\|F_n\|_{\mathcal{S}} \leq \|F\|_{\mathcal{S}}$, so the reverse inequality $L \leq \|F\|_{\mathcal{S}}$ follows. This proves the Fatou property.

Finally, fix $F \in \mathcal{S}$ and apply exactly the same extraction to a sequence of positive decompositions of $|F|$ whose total costs decrease to $\|F\|_{\mathcal{S}}$. Here the dominating function is simply $|F|$. The limiting four pieces have total norm at most $\|F\|_{\mathcal{S}}$, hence exactly that value by the definition of the infimum. Multiplication by the phase of F gives an attaining decomposition of F . $\square$

4 Verification and scope

The proof is computation-free. The following points were checked separately.

- The passage to positive decompositions loses no norm and gives the crucial domination $G_{i,n} \leq F$.
- The tensor $H = a \otimes b$ belongs to all four associate endpoint spaces, so (2) rules out an infinite-valued limit before the normalization by $1 + F$.
- Only four component norm budgets are present; a subsequence can make all of them converge, and Cesaro averaging preserves their limiting total.
- The simultaneous extraction occurs in one $\mathbb{R}^4$ -valued Hilbert space, so no incompatible componentwise subsequences are used.
- The argument does not address arbitrary infinite lattice sums. Its finite-component compactness is essential.

The bounded novelty check searched the run indexes, the exact sentence from the source remark, the source title and arXiv id, and close variants combining the four-corner sum with “Fatou property.” It also checked the metadata and scope of the Banach-lattice-sum paper cited by the source [2]. No later answer to this exact question was found as of 9 August 2026. Novelty confidence is moderate pending a specialist review.

Human review recommendation. Review the positive-decomposition lemma, the construction and use of the strictly positive dual tensor H , and the normalization/Cesaro extraction in Step 2. Those are the only nonformal points. If they pass, the proof gives a full affirmative answer to the source remark and the additional norm-attainment statement.

References

- [1] N. Holighaus and F. Voigtlaender, *Schur-type Banach modules of integral kernels acting on mixed-norm Lebesgue spaces*, J. Funct. Anal. 281 (2021), no. 9, 109197; arXiv:2006.01083v2.
- [2] M. Mastyło and E. A. Sánchez-Pérez, *Banach lattice sums with hereditary Fatou properties*, Funct. Approx. Comment. Math. 50 (2014), no. 2, 271–282.